

Chapter Contents

Chapter 31 - Introduction To Partial Differential Equations	4
Functions Of Several Variables	5
The Goal Of This Block	5
From A Curve To A Surface	5
First Partial Derivatives	6
Differentiate The Same Function With Respect To Y	6
Second And Mixed Derivatives	7
Check The Other Mixed Order	8
Reading Subscript Notation	8
Block 1 Summary	9
Classifying A Partial Differential Equation	9
The Goal Of This Block	9
The Order Comes From The Highest Derivative	9
Linear Partial Differential Equations	10
Semilinear, Quasilinear, And Fully Nonlinear	10
Worked Classification	11
Block 2 Summary	11
The Three Classical Model Equations	12
The Goal Of This Block	12
The One-Dimensional Heat Equation	12
The One-Dimensional Wave Equation	12
Laplace's Equation	13
Initial Data And Boundary Data	13
Second-Order Type In Two Variables	14
Classify The Three Models By Type	14
Block 3 Summary	15
Verifying Proposed Solutions	16
The Goal Of This Block	16
Worked Heat-Equation Verification	16
A Travelling-Wave Family	17
Differentiate The Travelling Wave	17
Worked Harmonic-Function Verification	18
A Fast Non-Solution Check	19
Block 4 Summary	20

Solving Simple PDEs By Integration	20
The Goal Of This Block	20
Why The Constant Becomes A Function	20
Worked First-Order Integration	21
Apply Data Along A Boundary	22
Integrate A Mixed-Derivative Equation	22
Verify The Mixed-Derivative Family	24
Block 5 Summary	24
Separating The Fixed-End Heat Equation	25
The Goal Of This Block	25
The Heat Problem	25
Substitute The Product	26
Why Both Ratios Must Be Constant	26
Obtain The Two Ordinary Differential Equations	27
Transfer The Boundary Conditions To X	28
Case One: Negative Lambda	29
Case Two: Zero Lambda	29
Case Three: Positive Lambda	31
Find The Allowed Spatial Frequencies	32
Solve The Temporal Equation	32
Block 6 Summary	33
Combining Heat Modes And Applying Initial Data	34
The Goal Of This Block	34
Why Modes May Be Added	34
A Finite Initial Temperature	34
Attach The Correct Time Factor To Each Mode	35
Verify The PDE	36
Verify The Boundary And Initial Conditions	37
General Initial Profiles	37
Block 7 Summary	38
Separating The Fixed-End Wave Equation	38
The Goal Of This Block	38
Separate The Wave Equation	38
Obtain The Spatial And Temporal Equations	39
Solve The Temporal Oscillator	40
Heat Decay Versus Wave Oscillation	41

Worked Wave Initial-Data Problem	42
Use The Initial Displacement	42
Use The Initial Velocity	43
Assemble And Check The Wave Solution	44
Block 8 Summary	45
A Reliable Introductory PDE Workflow	45
The Goal Of This Block	45
The Workflow	45
Common Mistake: Treating Other Variables As Numerical Con-	
stants Forever	46
Common Mistake: Confusing Two Kinds Of Separation	46
Common Mistake: Dividing Before Naming The Nontrivial Case	46
Common Mistake: Keeping Only The Convenient Sign	47
Common Mistake: Applying Superposition To A Nonlinear PDE	47
Common Mistake: Confusing Fixed And Insulated Ends	47
Block 9 Summary	48
Original Practice Set	48
Problems 1 To 4: Derivatives, Classification, And Verification	48
Problems 5 To 8: Harmonic Functions, Integration, And Separation	49
Problems 9 To 12: Heat And Wave Modes	50
Worked Answers: Problems 1 To 4	51
Worked Answers: Problems 5 To 8	54
Worked Answers: Problems 9 To 12	57
Chapter 31 Final Summary	61

Chapter 31 - Introduction To Partial Differential Equations

An ordinary differential equation describes an unknown function of one independent variable. A partial differential equation allows the unknown function to depend on several independent variables at once.

That change is more than a notation change. A solution now describes a surface, a temperature field, a vibrating string, or another distributed quantity. It may need initial data in time and boundary data in space.

This chapter will:

- explain partial derivatives and mixed derivatives in PDE notation
- classify equations by order and linearity
- introduce the heat, wave, and Laplace equations
- distinguish elliptic, parabolic, and hyperbolic second-order equations
- verify proposed solutions one derivative at a time
- integrate simple PDEs without losing arbitrary functions
- derive separated solutions of the fixed-end heat equation
- derive separated solutions of the fixed-end wave equation
- apply initial and boundary conditions to finite mode combinations
- build a reliable introductory PDE workflow

Functions Of Several Variables

The Goal Of This Block

The goal is to understand what a partial derivative measures and to make the subscript notation precise before using it inside differential equations.

a partial derivative changes one independent variable while temporarily holding the others fixed.

From A Curve To A Surface

An ODE commonly uses an unknown function such as:

$$y = y(x).$$

Its graph is a curve. A PDE may instead use:

$$u = u(x, y).$$

Its graph can be viewed as a surface above the xy -plane. The value $u(x, y)$ may represent, for example, temperature at the point (x, y) .

If time is also present, we may write:

$$u = u(x, y, t).$$

Here x and y are spatial variables and t is time. The number of independent variables is determined by the model, not by the letter used for the unknown function.

First Partial Derivatives

For $u = u(x, y)$, the derivative with respect to x is written:

$$u_x = \frac{\partial u}{\partial x}.$$

When calculating u_x , treat y as a constant. Similarly:

$$u_y = \frac{\partial u}{\partial y},$$

and x is held constant while differentiating with respect to y .

Consider:

$$u(x, y) = x^2y + e^{xy}.$$

Differentiate with respect to x , holding y fixed:

$$\begin{aligned} u_x &= \frac{\partial}{\partial x} (x^2y + e^{xy}) \\ &= 2xy + ye^{xy}. \end{aligned}$$

For the exponential term, the chain rule contributes $\partial(xy)/\partial x = y$.

Differentiate The Same Function With Respect To Y

Return to:

$$u(x, y) = x^2y + e^{xy}.$$

Differentiate with respect to y , holding x fixed:

$$\begin{aligned}u_y &= \frac{\partial}{\partial y} (x^2y + e^{xy}) \\&= x^2 + xe^{xy}.\end{aligned}$$

The two partial derivatives are different because they measure change in two different coordinate directions.

Second And Mixed Derivatives

Differentiate u_x again with respect to x to obtain:

$$u_{xx} = \frac{\partial^2 u}{\partial x^2}.$$

Differentiate u_x with respect to y to obtain the mixed derivative:

$$u_{xy} = \frac{\partial}{\partial y} (u_x) = \frac{\partial^2 u}{\partial y \partial x}.$$

For $u = x^2y + e^{xy}$, recall:

$$u_x = 2xy + ye^{xy}.$$

Differentiate this expression with respect to y :

$$\begin{aligned}u_{xy} &= \frac{\partial}{\partial y} (2xy + ye^{xy}) \\&= 2x + e^{xy} + xye^{xy}.\end{aligned}$$

The final two terms come from the product rule applied to ye^{xy} .

Check The Other Mixed Order

Recall:

$$u_y = x^2 + xe^{xy}.$$

Differentiate with respect to x :

$$\begin{aligned} u_{yx} &= \frac{\partial}{\partial x} (x^2 + xe^{xy}) \\ &= 2x + e^{xy} + xy e^{xy}. \end{aligned}$$

Thus:

$$u_{xy} = u_{yx}.$$

For functions with continuous mixed second derivatives, the order of these two different differentiations may be exchanged.¹

Reading Subscript Notation

For $u = u(x, t)$:

$$\begin{aligned} u_t &= \frac{\partial u}{\partial t}, \\ u_{xx} &= \frac{\partial^2 u}{\partial x^2}, \\ u_{xt} &= \frac{\partial}{\partial t} (u_x), \\ u_{xtt} &= \frac{\partial}{\partial t} \left(\frac{\partial}{\partial t} (u_x) \right). \end{aligned}$$

Read the derivative subscripts from left to right as the history of differentiation. Thus u_{xtt} means differentiate with respect to x once, then with respect to t twice.

¹ Equality of mixed partials requires regularity; it is not guaranteed for every imaginable function. Continuous second partial derivatives are a standard sufficient condition.

Block 1 Summary

A PDE solution depends on at least two independent variables. A partial derivative changes one variable at a time, and its subscript records the direction and order of differentiation.

Classifying A Partial Differential Equation

The Goal Of This Block

The goal is to classify PDEs by order and linearity without confusing the number of variables with the order of the equation.

The Order Comes From The Highest Derivative

The order of a PDE is the order of its highest derivative.

For example:

$$u_t + 3u_x = 0$$

is first order, because only first partial derivatives occur.

The equation:

$$u_{tt} - 4u_{xx} = 0$$

is second order. It involves two independent variables, but the number of variables does not determine the differential order.

The equation:

$$u_{xxy} + u_t = 0$$

is third order because u_{xxy} contains three differentiations.

Linear Partial Differential Equations

A PDE is linear in u when:

- u and all its derivatives occur only to the first power
- products such as uu_x or $u_x u_y$ do not occur
- coefficients may depend on the independent variables but not on u

For example:

$$(1 + x^2)u_{xx} + e^y u_y - 3u = \sin x$$

is linear. The coefficient $1 + x^2$ varies with x , and e^y varies with y , but neither coefficient depends on u .

The associated homogeneous equation is:

$$(1 + x^2)u_{xx} + e^y u_y - 3u = 0.$$

Semilinear, Quasilinear, And Fully Nonlinear

These terms describe how the highest-order derivatives appear.

A **semilinear** second-order equation is linear in the second derivatives, with coefficients depending only on the independent variables, but lower-order terms may be nonlinear. For example:

$$u_{xx} + u_{yy} + (u_x)^2 = 0.$$

A **quasilinear** equation is linear in its highest-order derivatives, but their coefficients may depend on u or lower derivatives. For example:

$$u_t + u u_x = 0$$

is first-order quasilinear: u_t and u_x occur linearly, but the coefficient of u_x is the unknown function u .

A **fully nonlinear** second-order equation contains nonlinear combinations of the highest derivatives. For example:

$$u_{xx}u_{yy} - (u_{xy})^2 = 1.$$

The classification describes algebraic structure; it does not say whether the equation is easy to solve.

Worked Classification

Classify:

$$xu_{tt} + u_x + u^3 = 0.$$

The highest derivative is u_{tt} , which is second order. Therefore the PDE is second order.

The highest derivative appears linearly, and its coefficient x depends only on an independent variable. The lower-order term u^3 is nonlinear.

The order and nonlinearity checks therefore give:

the equation is second-order semilinear.

Block 2 Summary

Order is determined by the highest derivative. Linearity asks how the unknown function and its derivatives occur, while semilinear and quasi-linear labels focus specifically on the highest-order derivatives.

The Three Classical Model Equations

The Goal Of This Block

The goal is to identify the heat, wave, and Laplace equations and connect each formula with the kind of behaviour it models.

The One-Dimensional Heat Equation

For a temperature $u(x, t)$ in a thin rod, the heat equation is:

$$u_t = \kappa u_{xx}, \quad \kappa > 0.$$

Here:

- x is position
- t is time
- $u(x, t)$ is temperature
- κ is thermal diffusivity

The second spatial derivative measures local curvature. At a local temperature maximum, $u_{xx} < 0$, so $u_t < 0$: the peak cools. At a local minimum, $u_{xx} > 0$, so $u_t > 0$: the dip warms.

In short:

heat diffusion smooths spatial differences as time passes.

The One-Dimensional Wave Equation

For the displacement $u(x, t)$ of a stretched string, the wave equation is:

$$u_{tt} = c^2 u_{xx}, \quad c > 0.$$

The second time derivative is acceleration, and the second spatial derivative measures curvature. The constant c is the wave speed.

Unlike the heat equation, the wave equation does not simply smooth a profile. It supports disturbances that travel and oscillate.

Laplace's Equation

For a steady field $u(x, y)$ in two spatial dimensions, Laplace's equation is:

$$u_{xx} + u_{yy} = 0.$$

The combination:

$$\nabla^2 u = u_{xx} + u_{yy}$$

is called the Laplacian. A function satisfying Laplace's equation is called **harmonic**.

Laplace's equation has no time variable. It commonly describes an equilibrium or steady state, such as a temperature distribution after transient changes have died away.²

Initial Data And Boundary Data

A time-dependent PDE usually needs data of two kinds.

An **initial condition** describes the state at one time. For example:

$$u(x, 0) = f(x).$$

A **boundary condition** describes what happens at the spatial edge. On $0 \leq x \leq L$, fixed zero endpoint values are:

$$u(0, t) = 0, \quad u(L, t) = 0.$$

² If a two-dimensional heat model satisfies $u_t = \kappa(u_{xx} + u_{yy})$, then a steady state has $u_t = 0$ and therefore satisfies $u_{xx} + u_{yy} = 0$.

For the wave equation, which contains a second time derivative, both initial displacement and initial velocity are commonly specified:

$$u(x, 0) = f(x), \quad u_t(x, 0) = g(x).$$

Second-Order Type In Two Variables

The second-order part of a linear PDE in variables x, y may be written:

$$Au_{xx} + 2Bu_{xy} + Cu_{yy}.$$

At a point, calculate:

$$\Delta = B^2 - AC.$$

The equation is classified there as:

$$\begin{array}{l|l} \Delta < 0 & \text{elliptic} \\ \Delta = 0 & \text{parabolic} \\ \Delta > 0 & \text{hyperbolic} \end{array}$$

This classification predicts broad qualitative behaviour; it does not replace the order and linearity classification.

Classify The Three Models By Type

For Laplace's equation:

$$u_{xx} + u_{yy} = 0,$$

For $u_{xx} + u_{yy} = 0$, the coefficients are $A = 1$, $B = 0$, and $C = 1$. Therefore:

$$\Delta = 0^2 - (1)(1) = -1 < 0,$$

so Laplace's equation is elliptic.

For the heat equation written as:

$$\kappa u_{xx} - u_t = 0,$$

in variables x, t , the second-order coefficients are $A = \kappa$, $B = 0$, and $C = 0$. Hence $\Delta = 0$, so it is parabolic.

For the wave equation written as:

$$c^2 u_{xx} - u_{tt} = 0,$$

we have $A = c^2$, $B = 0$, and $C = -1$. Thus:

$$\Delta = 0^2 - (c^2)(-1) = c^2 > 0,$$

so it is hyperbolic.

Block 3 Summary

Heat flow is parabolic, wave motion is hyperbolic, and Laplace equilibrium is elliptic. Their derivative structures reflect diffusion, propagation, and steady balance.

Verifying Proposed Solutions

The Goal Of This Block

The goal is to verify candidate solutions by calculating only the derivatives the PDE actually requires and then checking the resulting identity.

Worked Heat-Equation Verification

Verify that:

$$u(x, t) = e^{-9\kappa t} \sin(3x)$$

solves:

$$u_t = \kappa u_{xx}.$$

Differentiate with respect to t :

$$\begin{aligned} u_t &= \frac{\partial}{\partial t} (e^{-9\kappa t} \sin(3x)) \\ &= -9\kappa e^{-9\kappa t} \sin(3x). \end{aligned}$$

Now differentiate twice with respect to x :

$$\begin{aligned} u_x &= 3e^{-9\kappa t} \cos(3x), \\ u_{xx} &= -9e^{-9\kappa t} \sin(3x). \end{aligned}$$

Multiply u_{xx} by κ :

$$\kappa u_{xx} = -9\kappa e^{-9\kappa t} \sin(3x).$$

This is exactly u_t . Therefore the heat equation holds for every (x, t) in the domain.

A Travelling-Wave Family

Let F be twice differentiable and define:

$$u(x, t) = F(x - ct).$$

We will verify that every such function solves:

$$u_{tt} = c^2 u_{xx}.$$

Introduce the inner variable:

$$\tilde{\zeta} = x - ct.$$

Then:

$$u(x, t) = F(\tilde{\zeta}).$$

Keeping this inner variable visible prevents chain-rule factors from being lost.

Differentiate The Travelling Wave

Since $\tilde{\zeta} = x - ct$:

$$\tilde{\zeta}_x = 1, \quad \tilde{\zeta}_t = -c.$$

Differentiate with respect to x :

$$\begin{aligned} u_x &= F'(\tilde{\zeta})\tilde{\zeta}_x \\ &= F'(\tilde{\zeta}), \end{aligned}$$

Differentiate $u_x = F'(\xi)$ once more with respect to x :

$$\begin{aligned} u_{xx} &= F''(\xi)\xi_x \\ &= F''(\xi). \end{aligned}$$

Differentiate with respect to t :

$$\begin{aligned} u_t &= F'(\xi)\xi_t \\ &= -cF'(\xi). \end{aligned}$$

Differentiate once more with respect to t :

$$\begin{aligned} u_{tt} &= -cF''(\xi)\xi_t \\ &= -cF''(\xi)(-c) \\ &= c^2F''(\xi). \end{aligned}$$

Since $u_{xx} = F''(\xi)$:

$$u_{tt} = c^2u_{xx}.$$

Thus every twice-differentiable profile $F(x-ct)$ travels to the right at speed c without changing shape.

Worked Harmonic-Function Verification

Check whether:

$$u(x, y) = e^{2x} \cos(2y)$$

is harmonic.

Calculate the second x -derivative:

$$\begin{aligned}u_x &= 2e^{2x} \cos(2y), \\u_{xx} &= 4e^{2x} \cos(2y).\end{aligned}$$

Calculate the second y -derivative:

$$\begin{aligned}u_y &= -2e^{2x} \sin(2y), \\u_{yy} &= -4e^{2x} \cos(2y).\end{aligned}$$

Add the required derivatives:

$$\begin{aligned}u_{xx} + u_{yy} &= 4e^{2x} \cos(2y) - 4e^{2x} \cos(2y) \\&= 0.\end{aligned}$$

Therefore u satisfies Laplace's equation and is harmonic.

A Fast Non-Solution Check

Consider:

$$v(x, t) = e^{-\kappa t} \sin(2x).$$

For the heat equation $v_t = \kappa v_{xx}$:

$$v_t = -\kappa e^{-\kappa t} \sin(2x),$$

while:

$$\kappa v_{xx} = -4\kappa e^{-\kappa t} \sin(2x).$$

These expressions are not identical. Therefore v is not a solution, even though it has the same general product shape as a separated heat mode.

Block 4 Summary

Verification is substitution, not visual resemblance. Calculate the required partial derivatives, preserve every chain-rule factor, and compare both sides as functions of all independent variables.

Solving Simple PDEs By Integration

The Goal Of This Block

The goal is to integrate with respect to one variable while preserving the most general dependence on every variable held fixed.

in partial integration, the constant of integration becomes an arbitrary function of the untouched variables.

Why The Constant Becomes A Function

Suppose:

$$u_x = 0.$$

For each fixed y , the function does not change as x changes. Therefore u may still depend on y :

$$u(x, y) = F(y).$$

Writing only $u = C$ would lose valid solutions such as $u = y^2$ or $u = \sin y$. Both have $u_x = 0$.

For $u = u(x, y, z)$, the corresponding result is:

$$u_x = 0 \implies u = F(y, z).$$

Worked First-Order Integration

Find the general solution of:

$$u_x = 2x + 3y^2.$$

Integrate with respect to x , treating y as constant:

$$u(x, y) = \int (2x + 3y^2) dx.$$

Integrate each term:

$$\int 2x dx = x^2,$$

and, because $3y^2$ is constant with respect to x :

$$\int 3y^2 dx = 3xy^2.$$

Add the arbitrary function of the untouched variable:

$$u(x, y) = x^2 + 3xy^2 + F(y).$$

Check by differentiating with respect to x :

$$u_x = 2x + 3y^2 + 0.$$

Apply Data Along A Boundary

Now impose:

$$u(0, y) = e^y.$$

Start from the general family:

$$u(x, y) = x^2 + 3xy^2 + F(y).$$

Set $x = 0$:

$$\begin{aligned} u(0, y) &= 0^2 + 3(0)y^2 + F(y) \\ &= F(y). \end{aligned}$$

The boundary data require $u(0, y) = e^y$, so:

$$F(y) = e^y.$$

Substitute this function into the general family:

$$u(x, y) = x^2 + 3xy^2 + e^y.$$

Integrate A Mixed-Derivative Equation

Find the general solution of:

$$u_{xy} = 4xy.$$

To solve $u_{xy} = 4xy$, first integrate with respect to y . Since u_{xy} is the y -derivative of u_x :

$$u_x = \int 4xy \, dy.$$

Treat x as constant:

$$u_x = 2xy^2 + A(x).$$

The integration function is $A(x)$ because differentiating it with respect to y gives zero.

Now integrate with respect to x :

$$u = \int [2xy^2 + A(x)] \, dx.$$

Therefore:

$$u = x^2y^2 + G(x) + H(y),$$

where $G'(x) = A(x)$ and $H(y)$ is the new arbitrary function introduced by the second integration.

Hence:

$$\boxed{u(x, y) = x^2y^2 + G(x) + H(y).}$$

Verify The Mixed-Derivative Family

Start from:

$$u(x, y) = x^2y^2 + G(x) + H(y).$$

For $u = x^2y^2 + G(x) + H(y)$, differentiate first with respect to x :

$$u_x = 2xy^2 + G'(x).$$

Now differentiate with respect to y :

$$\begin{aligned}u_{xy} &= \frac{\partial}{\partial y} [2xy^2 + G'(x)] \\&= 4xy + 0 \\&= 4xy.\end{aligned}$$

Both arbitrary functions are valid: $G(x)$ disappears after the y derivative, and $H(y)$ disappeared during the first x derivative.

Block 5 Summary

When integrating with respect to one variable, preserve arbitrary dependence on all other variables. Repeated partial integration may produce more than one arbitrary function.

Separating The Fixed-End Heat Equation

The Goal Of This Block

The goal is to turn one PDE into two ODEs, justify the separation constant, and show why fixed endpoint conditions select only certain spatial frequencies.

The Heat Problem

Consider a rod on $0 \leq x \leq L$ whose endpoints are held at zero temperature:

$$u_t = \kappa u_{xx}, \quad \kappa > 0,$$

with:

$$u(0, t) = 0, \quad u(L, t) = 0.$$

The initial temperature will be imposed later. First we seek nonzero solutions that satisfy the PDE and the two boundary conditions.

Assume a product form:

$$u(x, t) = X(x)T(t).$$

This is an assumption about the form of a solution. It is called separation of variables because all x -dependence is placed in X and all t -dependence is placed in T .

Substitute The Product

From:

$$u(x, t) = X(x)T(t),$$

differentiate with respect to t :

$$u_t = X(x)T'(t).$$

Differentiate twice with respect to x :

$$u_{xx} = X''(x)T(t).$$

Substitute these expressions into $u_t = \kappa u_{xx}$:

$$X(x)T'(t) = \kappa X''(x)T(t).$$

For a nontrivial separated solution, work where $X(x)T(t) \neq 0$ and divide by $\kappa X(x)T(t)$:

$$\frac{T'(t)}{\kappa T(t)} = \frac{X''(x)}{X(x)}.$$

Why Both Ratios Must Be Constant

In:

$$\frac{T'(t)}{\kappa T(t)} = \frac{X''(x)}{X(x)},$$

the left side depends only on t , while the right side depends only on x . Yet the equality must hold for every admissible pair (x, t) .

Fix one value of t . The left side is then one number, so the right side must have that same value for every x . Therefore X''/X is constant. The same argument then makes $T'/(κT)$ that constant as well.

Write the common constant as $-\lambda$:

$$\frac{T'}{\kappa T} = \frac{X''}{X} = -\lambda.$$

The minus sign is a convention chosen because positive λ will produce the sine modes required by the fixed endpoints.

Obtain The Two Ordinary Differential Equations

Start with the spatial ratio:

$$\frac{X''}{X} = -\lambda.$$

Multiply by X :

$$X'' = -\lambda X.$$

Move every term to the left:

$$X'' + \lambda X = 0.$$

Now use the temporal ratio:

$$\frac{T'}{\kappa T} = -\lambda.$$

Multiply by κT :

$$T' + \kappa \lambda T = 0.$$

The PDE has become a spatial boundary-value problem and a temporal first-order ODE.

Transfer The Boundary Conditions To X

The first boundary condition is:

$$u(0, t) = 0.$$

Using $u = XT$:

$$X(0)T(t) = 0$$

for every t . A nontrivial separated solution cannot have $T(t)$ identically zero, so:

$$X(0) = 0.$$

Similarly, $u(L, t) = 0$ gives:

$$X(L)T(t) = 0,$$

and hence:

$$X(L) = 0.$$

The spatial problem is therefore:

$$X'' + \lambda X = 0, \quad X(0) = X(L) = 0.$$

Case One: Negative Lambda

Suppose $\lambda < 0$. Write:

$$\lambda = -\mu^2, \quad \mu > 0.$$

The spatial equation becomes:

$$X'' - \mu^2 X = 0.$$

Its general solution is:

$$X(x) = A \cosh(\mu x) + B \sinh(\mu x).$$

Apply $X(0) = 0$:

$$\begin{aligned} 0 &= A \cosh(0) + B \sinh(0) \\ &= A. \end{aligned}$$

Thus $A = 0$ and $X = B \sinh(\mu x)$. Now apply $X(L) = 0$:

$$B \sinh(\mu L) = 0.$$

Because $\mu L > 0$, $\sinh(\mu L) \neq 0$. Therefore $B = 0$, leaving only $X = 0$. Negative λ produces no nontrivial fixed-end mode.

Case Two: Zero Lambda

If $\lambda = 0$, the spatial equation is:

$$X'' = 0.$$

For the zero-separation case $X'' = 0$, integrate twice:

$$X(x) = A + Bx.$$

Apply $X(0) = 0$:

$$A = 0.$$

Then $X(L) = 0$ gives:

$$BL = 0.$$

Since $L > 0$, divide by L to obtain $B = 0$. Again only the trivial spatial solution remains.

Case Three: Positive Lambda

Suppose $\lambda > 0$ and define:

$$\alpha = \sqrt{\lambda}.$$

The spatial equation is:

$$X'' + \alpha^2 X = 0.$$

Its general solution is:

$$X(x) = A \cos(\alpha x) + B \sin(\alpha x).$$

Apply $X(0) = 0$:

$$\begin{aligned} 0 &= A \cos(0) + B \sin(0) \\ &= A. \end{aligned}$$

Therefore $A = 0$, so:

$$X(x) = B \sin(\alpha x).$$

Apply $X(L) = 0$:

$$B \sin(\alpha L) = 0.$$

For a nontrivial solution, $B \neq 0$. Divide by B :

$$\sin(\alpha L) = 0.$$

Find The Allowed Spatial Frequencies

The sine function is zero at integer multiples of π . Therefore:

$$\alpha L = n\pi, \quad n = 1, 2, 3, \dots$$

The value $n = 0$ would reproduce the already-trivial $\lambda = 0$ case, so the positive integers begin at 1.

Solve for α :

$$\alpha_n = \frac{n\pi}{L}.$$

Since $\lambda = \alpha^2$:

$$\lambda_n = \left(\frac{n\pi}{L} \right)^2.$$

The associated spatial functions are:

$$X_n(x) = \sin\left(\frac{n\pi x}{L}\right),$$

where an arbitrary nonzero multiplier has been left for the final solution.

Solve The Temporal Equation

For $\lambda = \lambda_n$, the temporal equation is:

$$T'_n + \kappa \lambda_n T_n = 0.$$

Insert $\lambda_n = (n\pi/L)^2$ into the temporal equation $T'_n + \kappa\lambda_n T_n = 0$:

$$T'_n + \kappa \left(\frac{n\pi}{L} \right)^2 T_n = 0.$$

This first-order linear ODE has solution:

$$T_n(t) = C_n e^{-\kappa(n\pi/L)^2 t}.$$

Multiply the spatial and temporal factors:

$$u_n(x, t) = C_n e^{-\kappa(n\pi/L)^2 t} \sin\left(\frac{n\pi x}{L}\right).$$

Each u_n is a separated heat mode.

Block 6 Summary

Fixed zero endpoints reject negative and zero separation values. Positive values are quantized as $(n\pi/L)^2$, and each spatial sine mode decays exponentially in time.

Combining Heat Modes And Applying Initial Data

The Goal Of This Block

The goal is to use linearity to combine separated modes and match a prescribed initial temperature without hiding where each time factor comes from.

Why Modes May Be Added

Write the heat operator as:

$$L[u] = u_t - \kappa u_{xx}.$$

It is linear. If $L[u_1] = 0$ and $L[u_2] = 0$, then for constants a, b :

$$\begin{aligned} L[au_1 + bu_2] &= aL[u_1] + bL[u_2] \\ &= 0. \end{aligned}$$

Therefore finite sums of heat modes are also solutions:

$$u(x, t) = \sum_{n=1}^N b_n e^{-\kappa(n\pi/L)^2 t} \sin\left(\frac{n\pi x}{L}\right).$$

The same statement extends to suitable infinite sums when convergence permits term-by-term differentiation.

A Finite Initial Temperature

On $0 \leq x \leq \pi$, solve:

$$u_t = 2u_{xx},$$

The remaining data for this heat problem are the fixed-end conditions:

$$u(0, t) = u(\pi, t) = 0,$$

and:

$$u(x, 0) = 4 \sin(2x) - 3 \sin(5x).$$

Here $L = \pi$ and $\kappa = 2$. The general n th mode is:

$$u_n = b_n e^{-2n^2 t} \sin(nx).$$

Only the frequencies $n = 2$ and $n = 5$ occur in the initial data.

Attach The Correct Time Factor To Each Mode

For $n = 2$, the decay exponent is:

$$-2n^2 t = -2(2^2)t = -8t.$$

Thus the first mode is:

$$4e^{-8t} \sin(2x).$$

For $n = 5$:

$$-2n^2 t = -2(5^2)t = -50t.$$

For $n = 5$, the coefficient is -3 and the time factor is e^{-50t} , so the second mode is:

$$-3e^{-50t} \sin(5x).$$

Combine them:

$$u(x, t) = 4e^{-8t} \sin(2x) - 3e^{-50t} \sin(5x).$$

Verify The PDE

Start from:

$$u = 4e^{-8t} \sin(2x) - 3e^{-50t} \sin(5x).$$

Differentiate with respect to t :

$$u_t = -32e^{-8t} \sin(2x) + 150e^{-50t} \sin(5x).$$

Differentiate twice with respect to x :

$$u_{xx} = -16e^{-8t} \sin(2x) + 75e^{-50t} \sin(5x).$$

Multiply by 2:

$$2u_{xx} = -32e^{-8t} \sin(2x) + 150e^{-50t} \sin(5x).$$

This is exactly u_t , so the PDE is satisfied.

Verify The Boundary And Initial Conditions

At $x = 0$, both sine factors vanish:

$$u(0, t) = 0.$$

At $x = \pi$:

$$\sin(2\pi) = 0, \quad \sin(5\pi) = 0,$$

so:

$$u(\pi, t) = 0.$$

Set $t = 0$. Since $e^0 = 1$:

$$\begin{aligned} u(x, 0) &= 4e^0 \sin(2x) - 3e^0 \sin(5x) \\ &= 4 \sin(2x) - 3 \sin(5x). \end{aligned}$$

The candidate satisfies the PDE and all three conditions.

General Initial Profiles

For a more general initial temperature $f(x)$, seek:

$$u(x, t) = \sum_{n=1}^{\infty} b_n e^{-\kappa(n\pi/L)^2 t} \sin\left(\frac{n\pi x}{L}\right).$$

At $t = 0$:

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right).$$

To determine the coefficients b_n , use the orthogonality of the sine functions on $[0, L]$. Their projection coefficients are:

$$b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx.$$

This is the bridge from separation of variables to Fourier sine series.

Block 7 Summary

Linearity allows separated heat modes to be superposed. The initial profile determines their coefficients, while higher spatial frequencies decay faster because their exponents contain n^2 .

Separating The Fixed-End Wave Equation

The Goal Of This Block

The goal is to reuse the spatial eigenfunctions from the heat problem while showing why the wave equation produces time oscillation instead of decay.

Separate The Wave Equation

Consider:

$$u_{tt} = c^2 u_{xx},$$

with fixed endpoints:

$$u(0, t) = u(L, t) = 0.$$

For this fixed-end wave equation, seek a separated solution:

$$u(x, t) = X(x)T(t).$$

Its required second derivatives are:

$$u_{tt} = X(x)T''(t), \quad u_{xx} = X''(x)T(t).$$

Substitute into the wave equation:

$$X(x)T''(t) = c^2X''(x)T(t).$$

Divide a nontrivial product by c^2XT :

$$\frac{T''}{c^2T} = \frac{X''}{X} = -\lambda.$$

Obtain The Spatial And Temporal Equations

From:

$$\frac{X''}{X} = -\lambda,$$

we obtain:

$$X'' + \lambda X = 0.$$

The fixed endpoint conditions again give:

$$X(0) = X(L) = 0.$$

This is the same fixed-end spatial problem analysed in Block 6. Its non-trivial solutions therefore occur at:

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad X_n(x) = \sin\left(\frac{n\pi x}{L}\right).$$

From the temporal ratio:

$$\frac{T''}{c^2 T} = -\lambda_n,$$

multiply by $c^2 T$:

$$T_n'' + c^2 \lambda_n T_n = 0.$$

Solve The Temporal Oscillator

Substitute:

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2$$

into:

$$T_n'' + c^2 \lambda_n T_n = 0.$$

This gives:

$$T_n'' + \left(\frac{cn\pi}{L}\right)^2 T_n = 0.$$

Define the angular frequency:

$$\omega_n = \frac{cn\pi}{L}.$$

Then:

$$T_n(t) = A_n \cos(\omega_n t) + B_n \sin(\omega_n t).$$

The n th fixed-end wave mode is:

$$u_n(x, t) = [A_n \cos(\omega_n t) + B_n \sin(\omega_n t)] \sin\left(\frac{n\pi x}{L}\right).$$

Heat Decay Versus Wave Oscillation

The heat mode has time factor:

$$e^{-\kappa(n\pi/L)^2 t}.$$

It decreases toward zero.

The wave mode has time factors:

$$\cos\left(\frac{cn\pi t}{L}\right), \quad \sin\left(\frac{cn\pi t}{L}\right).$$

They oscillate without decay in the ideal model.

The same boundary conditions produce the same spatial sine functions, but the different time derivatives in the PDE produce different temporal behaviour.

Worked Wave Initial-Data Problem

On $0 \leq x \leq \pi$, let $c = 2$ and impose:

$$u(0, t) = u(\pi, t) = 0,$$

with initial displacement:

$$u(x, 0) = 2 \sin x - \sin(3x),$$

and initial velocity:

$$u_t(x, 0) = 4 \sin(2x).$$

Since $L = \pi$ and $c = 2$:

$$\omega_n = \frac{cn\pi}{L} = 2n.$$

Only modes $n = 1, 2, 3$ are needed.

Use The Initial Displacement

Write the relevant finite sum:

$$u(x, t) = \sum_{n=1}^3 [A_n \cos(2nt) + B_n \sin(2nt)] \sin(nx).$$

Set $t = 0$. Since $\cos 0 = 1$ and $\sin 0 = 0$:

$$u(x, 0) = A_1 \sin x + A_2 \sin(2x) + A_3 \sin(3x).$$

Match this displacement expansion with the prescribed initial displacement:

$$u(x, 0) = 2 \sin x - \sin(3x).$$

Comparing the sine coefficients therefore gives:

$$A_1 = 2, \quad A_2 = 0, \quad A_3 = -1.$$

Use The Initial Velocity

Differentiate the finite mode sum with respect to t :

$$u_t(x, t) = \sum_{n=1}^3 [-2nA_n \sin(2nt) + 2nB_n \cos(2nt)] \sin(nx).$$

Set $t = 0$:

$$u_t(x, 0) = 2B_1 \sin x + 4B_2 \sin(2x) + 6B_3 \sin(3x).$$

Match this with the prescribed velocity:

$$u_t(x, 0) = 4 \sin(2x).$$

Compare the sine coefficients:

$$2B_1 = 0, \quad 4B_2 = 4, \quad 6B_3 = 0.$$

Solving these three coefficient equations gives:

$$B_1 = 0, \quad B_2 = 1, \quad B_3 = 0.$$

Assemble And Check The Wave Solution

Insert the calculated constants into the mode sum:

$$u(x, t) = 2 \cos(2t) \sin x + \sin(4t) \sin(2x) - \cos(6t) \sin(3x).$$

Every spatial factor vanishes at $x = 0$ and $x = \pi$, so both endpoint conditions hold.

At $t = 0$:

$$u(x, 0) = 2 \sin x - \sin(3x).$$

Differentiate the final expression:

$$u_t = -4 \sin(2t) \sin x + 4 \cos(4t) \sin(2x) + 6 \sin(6t) \sin(3x).$$

Set $t = 0$:

$$u_t(x, 0) = 4 \sin(2x).$$

Each term is a separated wave mode, so linearity ensures the sum satisfies the wave equation.

Block 8 Summary

The fixed-end heat and wave equations share spatial sine eigenfunctions. Heat modes decay exponentially, whereas wave modes oscillate and require both initial displacement and initial velocity.

A Reliable Introductory PDE Workflow

The Goal Of This Block

The goal is to turn the chapter into a dependable sequence and identify the mistakes that most often hide inside compact PDE notation.

The Workflow

Step 1: List the variables. State the dependent variable and every independent variable.

Step 2: Classify the equation. Identify order, linearity, and, when useful, elliptic/parabolic/hyperbolic type.

Step 3: List all conditions. Separate initial data from spatial boundary data.

Step 4: If verifying, calculate only the derivatives required by the PDE. Carry each source expression into its derivative.

Step 5: If integrating, name the held-fixed variables. Add an arbitrary function of those variables.

Step 6: If separating, write $u = XT$ and substitute before dividing. Do not state separated ODEs without showing the ratio equation.

Step 7: Analyze every sign of the separation constant. Let the boundary conditions decide which sign produces nontrivial modes.

Step 8: Solve the temporal equation only after the spatial values are known. This keeps the mode index and time factor connected.

Step 9: Apply each condition locally. Restate the current solution before substitution.

Step 10: Verify the final PDE, boundary data, and initial data separately.

Common Mistake: Treating Other Variables As Numerical Constants Forever

When integrating u_x , the variable y is held constant during the integration, but the result may still depend on y afterward.

Thus:

$$u_x = 2x$$

has general solution:

$$u = x^2 + F(y),$$

not merely $u = x^2 + C$.

Common Mistake: Confusing Two Kinds Of Separation

For a separable ODE, variables are rearranged into expressions such as:

$$g(y) dy = f(x) dx.$$

For PDE separation of variables, we assume a product:

$$u(x, t) = X(x)T(t).$$

The methods share a name but perform different operations.

Common Mistake: Dividing Before Naming The Nontrivial Case

The step:

$$XT' = \kappa X''T \implies \frac{T'}{\kappa T} = \frac{X''}{X}$$

divides by X and T . State that the calculation seeks a nontrivial separated solution and works on regions where the factors are nonzero. The resulting ODE solutions extend across their isolated zeros.

Common Mistake: Keeping Only The Convenient Sign

The fixed-end heat derivation eventually uses $\lambda > 0$, but that conclusion comes from testing:

$$\lambda < 0, \quad \lambda = 0, \quad \lambda > 0.$$

Skipping the first two cases hides why positive discrete values are selected.

Common Mistake: Applying Superposition To A Nonlinear PDE

If u_1 and u_2 solve a linear homogeneous PDE, then $au_1 + bu_2$ also solves it. This does not generally hold for nonlinear equations.

For example, the product term in:

$$u_t + uu_x = 0$$

creates cross terms when two solutions are added. Check linearity before building a mode sum.

Common Mistake: Confusing Fixed And Insulated Ends

Fixed zero temperature uses value conditions:

$$u(0, t) = u(L, t) = 0.$$

Insulated ends use zero heat-flux conditions, which in the simplest model are:

$$u_x(0, t) = u_x(L, t) = 0.$$

These conditions select different spatial eigenfunctions. Read the physical condition before writing the mathematics.

Block 9 Summary

PDE work is reliable when variables, derivative directions, arbitrary functions, separation signs, domains, and all initial and boundary conditions remain explicit.

Original Practice Set

Problems 1 To 4: Derivatives, Classification, And Verification

Problem 1: Calculate Partial Derivatives. For:

$$u(x, y) = xy^2 + \cos(xy),$$

find u_x , u_y , and u_{xy} .

Problem 2: Classify A PDE. Classify by order and linearity:

$$u_{xx} + u_{tt} + e^u = 0.$$

Problem 3: Identify The Classical Type. Classify each as elliptic, parabolic, or hyperbolic:

$$u_{xx} + 4u_{yy} = 0,$$

$$u_t - 5u_{xx} = 0,$$

$$9u_{xx} - u_{tt} = 0.$$

Problem 4: Verify A Heat Solution. Verify that:

$$u(x, t) = e^{-16\kappa t} \cos(4x)$$

satisfies $u_t = \kappa u_{xx}$.

Problems 5 To 8: Harmonic Functions, Integration, And Separation

Problem 5: Check A Harmonic Polynomial. Determine whether:

$$u(x, y) = x^3 - 3xy^2$$

satisfies Laplace's equation.

Problem 6: Integrate And Apply Data. Solve:

$$u_x = 4x - y, \quad u(0, y) = y^2.$$

Problem 7: Integrate A Mixed Derivative. Find the general solution of:

$$u_{xy} = 6x^2y.$$

Problem 8: Separate A Heat Equation. For:

$$u_t = 3u_{xx}, \quad u(0, t) = u(L, t) = 0,$$

assume $u = XT$ and derive the two ODEs after setting the common ratio to $-\lambda$.

Problems 9 To 12: Heat And Wave Modes

Problem 9: Find Fixed-End Eigenvalues. Solve:

$$X'' + \lambda X = 0, \quad X(0) = X(2) = 0$$

for its positive eigenvalues and corresponding sine eigenfunctions.

Problem 10: Solve A Finite Heat Problem. On $0 \leq x \leq \pi$, solve:

$$u_t = 3u_{xx}, \quad u(0, t) = u(\pi, t) = 0,$$

with:

$$u(x, 0) = 2 \sin x + 5 \sin(4x).$$

Problem 11: Verify A Wave Mode. Verify that:

$$u(x, t) = \sin\left(\frac{2\pi x}{L}\right) \cos\left(\frac{2\pi ct}{L}\right)$$

satisfies the wave equation and fixed endpoint conditions.

Problem 12: Match Wave Initial Data. On $0 \leq x \leq \pi$ with $c = 1$, find the fixed-end wave solution with:

$$u(x, 0) = 3 \sin(2x), \quad u_t(x, 0) = -4 \sin(4x).$$

Worked Answers: Problems 1 To 4

Answer 1. Use:

$$u(x, y) = xy^2 + \cos(xy).$$

Differentiate with respect to x , holding y fixed:

$$\begin{aligned} u_x &= \frac{\partial}{\partial x} [xy^2 + \cos(xy)] \\ &= y^2 - y \sin(xy). \end{aligned}$$

Differentiate with respect to y , holding x fixed:

$$\begin{aligned} u_y &= \frac{\partial}{\partial y} [xy^2 + \cos(xy)] \\ &= 2xy - x \sin(xy). \end{aligned}$$

To find u_{xy} , differentiate u_x with respect to y :

$$\begin{aligned} u_{xy} &= \frac{\partial}{\partial y} [y^2 - y \sin(xy)] \\ &= 2y - \sin(xy) - xy \cos(xy). \end{aligned}$$

Answer 2. The equation is:

$$u_{xx} + u_{tt} + e^u = 0.$$

The highest derivatives, u_{xx} and u_{tt} , are second order and occur linearly with constant coefficients. The lower-order term e^u is nonlinear. The order and nonlinearity checks therefore give:

the PDE is second-order semilinear.

Answer 3. For:

$$u_{xx} + 4u_{yy} = 0,$$

$A = 1, B = 0$, and $C = 4$. Thus:

$$B^2 - AC = 0 - (1)(4) = -4 < 0,$$

so the equation is elliptic.

For:

$$u_t - 5u_{xx} = 0,$$

the second-order coefficients in (x, t) are $A = -5, B = 0$, and $C = 0$. Thus $B^2 - AC = 0$, so the equation is parabolic.

For:

$$9u_{xx} - u_{tt} = 0,$$

$A = 9, B = 0$, and $C = -1$. Therefore:

$$B^2 - AC = 0 - (9)(-1) = 9 > 0,$$

so the equation is hyperbolic.

Answer 4. Start with:

$$u(x, t) = e^{-16\kappa t} \cos(4x).$$

For the candidate $u = e^{-16\kappa t} \cos(4x)$, differentiate with respect to t :

$$u_t = -16\kappa e^{-16\kappa t} \cos(4x).$$

Differentiate twice with respect to x :

$$\begin{aligned}u_x &= -4e^{-16\kappa t} \sin(4x), \\u_{xx} &= -16e^{-16\kappa t} \cos(4x).\end{aligned}$$

Multiply by κ :

$$\kappa u_{xx} = -16\kappa e^{-16\kappa t} \cos(4x) = u_t.$$

Therefore the heat equation is satisfied.

Worked Answers: Problems 5 To 8**Answer 5.** For:

$$u(x, y) = x^3 - 3xy^2,$$

calculate:

$$u_x = 3x^2 - 3y^2,$$

$$u_{xx} = 6x.$$

Differentiate with respect to y :

$$u_y = -6xy,$$

$$u_{yy} = -6x.$$

Add:

$$u_{xx} + u_{yy} = 6x - 6x = 0.$$

Therefore the polynomial is harmonic.

Answer 6. Integrate:

$$u_x = 4x - y$$

with respect to x , treating y as constant:

$$u(x, y) = 2x^2 - xy + F(y).$$

Apply the boundary data $u(0, y) = y^2$ to $u = 2x^2 - xy + F(y)$:

$$\begin{aligned}u(0, y) &= 2(0)^2 - (0)y + F(y) \\&= F(y).\end{aligned}$$

Therefore $F(y) = y^2$, and:

$$u(x, y) = 2x^2 - xy + y^2.$$

Answer 7. Start from:

$$u_{xy} = 6x^2y.$$

Integrate with respect to y :

$$u_x = 3x^2y^2 + A(x).$$

Integrate with respect to x :

$$\begin{aligned}u(x, y) &= \int [3x^2y^2 + A(x)] dx \\&= x^3y^2 + G(x) + H(y),\end{aligned}$$

where $G'(x) = A(x)$. Hence:

$$u(x, y) = x^3y^2 + G(x) + H(y).$$

Answer 8. For the heat equation $u_t = 3u_{xx}$, seek a separated solution:

$$u(x, t) = X(x)T(t).$$

Then:

$$u_t = XT', \quad u_{xx} = X''T.$$

Substitute into $u_t = 3u_{xx}$:

$$XT' = 3X''T.$$

Divide a nontrivial product by $3XT$:

$$\frac{T'}{3T} = \frac{X''}{X} = -\lambda.$$

From $X''/X = -\lambda$:

$$\boxed{X'' + \lambda X = 0.}$$

From $T'/(3T) = -\lambda$:

$$\boxed{T' + 3\lambda T = 0.}$$

Worked Answers: Problems 9 To 12

Answer 9. For positive λ , write $\alpha = \sqrt{\lambda}$. Then:

$$X(x) = A \cos(\alpha x) + B \sin(\alpha x).$$

Apply $X(0) = 0$:

$$A = 0.$$

Thus $X = B \sin(\alpha x)$. Apply $X(2) = 0$:

$$B \sin(2\alpha) = 0.$$

For a nontrivial eigenfunction, $B \neq 0$, so:

$$\sin(2\alpha) = 0.$$

Therefore:

$$2\alpha = n\pi, \quad n = 1, 2, 3, \dots$$

Divide by 2 and square:

$$\lambda_n = \left(\frac{n\pi}{2}\right)^2.$$

The corresponding eigenfunctions may be chosen as:

$$X_n(x) = \sin\left(\frac{n\pi x}{2}\right).$$

Answer 10. For $L = \pi$ and $\kappa = 3$, the n th heat mode is:

$$b_n e^{-3n^2 t} \sin(nx).$$

The initial profile contains modes $n = 1$ and $n = 4$.

For $n = 1$:

$$-3n^2 t = -3t.$$

For $n = 4$:

$$-3n^2 t = -3(16)t = -48t.$$

Attach these factors to the initial coefficients:

$$u(x, t) = 2e^{-3t} \sin x + 5e^{-48t} \sin(4x).$$

At $t = 0$, both exponentials equal 1, recovering the prescribed initial temperature.

Answer 11. Use:

$$u(x, t) = \sin\left(\frac{2\pi x}{L}\right) \cos\left(\frac{2\pi c t}{L}\right).$$

Let:

$$a = \frac{2\pi}{L}.$$

With $a = 2\pi/L$, the proposed function becomes $u = \sin(ax) \cos(akt)$. Differentiate it twice in time:

$$u_{tt} = -a^2 c^2 \sin(ax) \cos(akt).$$

Differentiate twice in space:

$$u_{xx} = -a^2 \sin(ax) \cos(akt).$$

Multiply by c^2 :

$$c^2 u_{xx} = -a^2 c^2 \sin(ax) \cos(akt) = u_{tt}.$$

At $x = 0$:

$$\sin(0) = 0.$$

At $x = L$:

$$\sin(aL) = \sin(2\pi) = 0.$$

Thus the PDE and both fixed endpoint conditions hold.

Answer 12. For $L = \pi$ and $c = 1$, the n th wave frequency is:

$$\omega_n = n.$$

The displacement contains mode $n = 2$, and the velocity contains mode $n = 4$. Use:

$$u(x, t) = A_2 \cos(2t) \sin(2x) + B_4 \sin(4t) \sin(4x).$$

Apply the initial displacement at $t = 0$ to this two-mode wave sum:

$$u(x, 0) = A_2 \sin(2x).$$

Match $u(x, 0) = 3 \sin(2x)$:

$$A_2 = 3.$$

Differentiate the mode sum:

$$u_t = -2A_2 \sin(2t) \sin(2x) + 4B_4 \cos(4t) \sin(4x).$$

Set $t = 0$:

$$u_t(x, 0) = 4B_4 \sin(4x).$$

Match $u_t(x, 0) = -4 \sin(4x)$:

$$4B_4 = -4, \quad B_4 = -1.$$

Therefore:

$$u(x, t) = 3 \cos(2t) \sin(2x) - \sin(4t) \sin(4x).$$

Chapter 31 Final Summary

A PDE contains partial derivatives of a function with several independent variables. The three central models are:

$$\begin{aligned} u_t &= \kappa u_{xx} && \text{heat,} \\ u_{tt} &= c^2 u_{xx} && \text{wave,} \\ u_{xx} + u_{yy} &= 0 && \text{Laplace.} \end{aligned}$$

When integrating a partial derivative, preserve arbitrary dependence on every untouched variable:

$$u_x = f(x, y) \implies u = \int f(x, y) dx + F(y).$$

For fixed zero endpoints on $[0, L]$, separation of variables selects:

$$\lambda_n = \left(\frac{n\pi}{L} \right)^2, \quad X_n(x) = \sin \left(\frac{n\pi x}{L} \right).$$

The same spatial modes have different time factors:

$$\begin{aligned} \text{heat:} \quad & e^{-\kappa(n\pi/L)^2 t}, \\ \text{wave:} \quad & A_n \cos \left(\frac{cn\pi t}{L} \right) + B_n \sin \left(\frac{cn\pi t}{L} \right). \end{aligned}$$

partial derivatives describe change in several directions, separation turns the PDE into linked ODEs, boundary conditions select the spatial modes, and initial data determine how those modes are combined.